

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B.TECH DEGREE SECOND SEMESTER FIRST INTERNAL EXAMINATION
CE A /CE B /ME B
23-200-0205A BASIC ELECTRONICS ENGINEERING
(2023 scheme)

Time : 2 Hours

Max. Marks: 30

Course Outcomes:

On successful completion of this course the student will be able to:

CO1 : Illustrate the concept of diode in rectifiers, filter circuits and wave shaping,

CO2 : Interpret the functioning of oscillators and operational amplifiers.

CO3 : Explain the principle of embedded systems and sensors

CO4 : Summarise the functioning of a communication system and different modulation technologies

Bloom's Taxonomy Levels (BL) : L1- Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate,
L6 - Create

PO - Programme Outcome

PART – A

(Answer all questions.)

	(5 x 2 =10)	Marks	BL	CO	PO
I (a) What is doping ? How p-type semiconductors are formed?	2	1	1	1	
(b) What are rectifiers? Which are the different types of rectifiers?	2	2	1	1	
(c) Mention the Barkhausen's criteria for sustained oscillations	2	1	2	1	
(d) What do you mean by the gain of an amplifier?	2	2	1	1	
(e) Define the cut-off frequencies in the frequency response of an amplifier	2	2	1	1	

(P.T.O)

PART – B

	(2 x 10 = 20)	Marks	BL	CO	PO
II (a) Explain the working of a full wave bridge rectifier with circuit diagram and waveforms.		7	2	1	1,5
(b). Draw the VI characteristics of a zener diode		3	2	1	1
(OR)					
III (a) What are Amplifiers ? Which are the different types of amplifiers		5	1	1	1,5
(b) Explain the frequency response of an amplifier with suitable diagrams.		5	2	1	1
IV (a) Explain the working of a half wave bridge rectifier with circuit diagram and waveforms.		7	2	1	1,5
(b) An amplifier with negative feedback applied has an open-loop gain of 50 and one tenth of its output is fed back to the input (feedback ratio = 0.1). Determine the overall voltage gain.		3	3	1	1
(OR)					
V (a) What are oscillators? Derive the overall voltage gain for an amplifier with positive feedback.		6	2	2	1,5
(b) An amplifier produces an output voltage of 4V for an input of 50mV. If the input and output currents in this conditions are 4mA and 100mA respectively, determine the voltage gain, current gain and power gain of the amplifier.		4	3	1	1,5
